SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	**HRISSA**
Selected Domain	**Domain 2**
Selected SDG Track (1–17)	**SDG 8 – Decent work and Economic growth**
Project / Solution Name	**Giser.AI**
Name (Leader)	**Azer Hamdaoui**
Email & Phone (Leader)	**azer.hamdaoui.07@gmail.com +216 56138490(what'sapp)**
University Affiliation(s)	**Xiamen University Malaysia**
Number of Team Members	**5**

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

Many small and medium-sized enterprises (SMEs) and service-based startups in Malaysia face a common yet often overlooked challenge: the time-consuming back-and-forth communication required to understand customer requests. Managers frequently need to review WhatsApp messages, voice notes, and images, clarify measurements and job requirements, respond to inquiries, and discuss costs before they can begin planning a job. For example, a small painting business may need to exchange multiple messages with a customer just to gather sufficient information to prepare a quotation. With limited staff and resources, these repetitive administrative tasks can quickly overwhelm managers and divert their attention from essential business operations. Economically, the time spent on repetitive communication reduces productivity, delays quotations, increases the risk of losing potential customers, and limits business growth, ultimately affecting profitability and competitiveness. Socially, constant interruptions and excessive workloads contribute to managerial stress and burnout, while delayed responses and uncertainty can weaken customer trust and satisfaction. Environmentally, incomplete or inaccurate job information may lead to incorrect material estimates, unnecessary purchases, excess paint and material waste, and avoidable trips to suppliers or job sites. These impacts are interconnected, as inefficient communication not only consumes valuable time and human effort but also affects business performance, employee well-being, customer experience, and resource utilisation. Therefore, addressing this challenge is essential to helping Malaysian SMEs streamline their operations, improve productivity, support a healthier working environment, deliver more reliable customer service, and adopt more sustainable business practices.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

Giser.AI is an AI-powered application designed to help Malaysian SMEs, particularly service-based businesses, streamline customer communication and job preparation. By reducing repetitive inquiries and manual information gathering, the application enables business owners and managers to focus on service delivery, improve productivity, and respond to customers more efficiently.

Giser.AI applies Generative AI (GenAI) to understand customer messages, voice notes, and images, transforming unstructured information into structured, actionable job briefs. Using GenAI APIs and prompt engineering, the system extracts essential job requirements, such as measurements, service types, and customer preferences, while identifying missing information that requires clarification. Through planned WhatsApp integration, the application aims to streamline the collection of customer information within familiar communication channels.

Based on the information gathered, Giser.AI is designed to generate job summaries, estimate costs and required materials, and identify suitable time slots. Managers can then review the generated information and accept or decline jobs with a single click, reducing administrative workload and improving quotation turnaround times.

Building on our working prototype, we intend to leverage technical skills gained from the SDC Phase 1 workshops, including GenAI API integration, prompt engineering, database architecture, cloud hosting, and AI agent or workflow automation. These skills will support the development of reliable AI workflows, structured data storage, and scalable deployment. We are also exploring cloud-based development and hosting options, including Replit and RunPod, based on the application's computational and deployment requirements.

Ultimately, Giser.AI aims to help Malaysian SMEs operate more efficiently by reducing communication delays, improving customer satisfaction, minimising estimation errors, and supporting more responsible use of materials and resources.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

(appendix attached at the end)

1. Communication & Frontend Interfaces

- **WhatsApp Business Cloud API:** The main channel where clients chat with us, share their room photos, and lay out their painting needs.
- **Vapi:** Handles inbound customer phone calls smoothly, taking care of speech-to-text, natural voice responses, and awkward mid-sentence interruptions.
- **Contractor Web Dashboard:** A clean Replit-hosted portal where painters can quickly look over auto-generated estimates, tweak pricing if needed, and confirm schedule slots.

2. Backend & Core Frameworks

- **FastAPI:** The engine connecting everything behind the scenes—handling webhooks, routing AI requests, and running our core calculation logic.
- **ReportLab:** Turns the finalized project numbers into a clean, professional PDF quote that can be sent straight to the client.
- **Uvicorn:** Runs our FastAPI services reliably with minimal overhead.

3. AI & Compute Infrastructure

- **RunPod:** Gives us dedicated GPU power to run our open-source LLM specifically for pulling structured measurements and job specs out of messy chat messages.
- **PyTorch & Hugging Face:** The runtime stack that loads and serves our extraction model efficiently.

4. External Services & Productivity APIs

- **Google Calendar API (v3):** Keeps bookings synced in real time to prevent double-booking and lock in agreed-upon project dates automatically.

5. Database & State Management

- **SQL Database (SQLite / PostgreSQL):** Keeps a reliable log of chat histories, contractor rate cards, draft estimates, and customer booking details.
- **Replit Cloud:** The live hosting environment running our backend APIs, contractor dashboard, and database setup in one place.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

 RunPod (Cloud GPU Infrastructure): Hosts the GenAI LLM to extract structured job parameters (e.g., room dimensions, surface conditions) while keeping conversational interpretation isolated from pricing and scheduling logic.

 Vapi (Voice AI Platform): Powers conversational voice interactions by managing speech recognition, text-to-speech, interruption handling, and natural audio dialogue.

 Replit (Cloud Hosting Platform): Hosts the core FastAPI backend and the SME dashboard for reviewing, editing, and approving quotes and appointments.

 WhatsApp Business Cloud API (Messaging Interface): Provides the customer-facing chat channel to receive project requirements and job photos.

 Google Calendar API (v3) (Cloud Integration): Automates real-time availability scans, checks for double-booking conflicts, and syncs confirmed appointments.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

Our team applied **design-thinking methodology** to develop Giser.ai through three key steps. **First, we identified** the challenges faced by painting SMEs, including scattered WhatsApp enquiries, time-consuming quotations, and missed follow-ups. **Second, we designed and prototyped** an AI assistant that extracts job details, identifies missing information, and prepares quotation drafts using predefined pricing rules and human approval. **Finally, we plan to test and improve** the MVP using realistic scenarios and feedback from potential users to evaluate accuracy, usability, and processing time. By Phase 2, our goal is to deliver a technically viable, user-friendly solution with appropriate privacy safeguards, verified integration requirements, and a clear deployment plan for real-world use.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess [Innovation and Creativity as well as Feasibility of build]. Vague or unsupported projections may weaken your case for advancing to Phase 2.

Metric Category	**Your Response**
Target SDG Sub-Target(s)	SDG 8: Decent work and economic growth SDG 8.1: Economic Growth SDG8.2 Diversification and Innovation SDG 8.3: Enterprise Support
Metric Category	**Your Response**
Identify the exact UN SDG sub-target your project addresses *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	- Gesir.AI Targets 8.2 (Diversification and Innovation) which Boost productivity via technological upgrades, innovation, and diversification into high-value and labor-intensive sectors.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Statistics show: - Small business owners typically spend **10 to 15 hours per week** answering support queries themselves, which translates to roughly $19,500 per year in lost productivity (Time burden) (Ringly.io,2026) - Small business owners typically spend **10 to 15 hours per week** answering support queries themselves, which translates to roughly $19,500 per year in lost productivity (Budget Allocation) (Ringly.io,2026)
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	The amount an SME spends varies significantly by how they handle communication: • **Phone support:** $6 to $8 per interaction • **Email and chat:** $3 to $5 per interaction • **AI voice/chat agents:** Roughly $0.40 per call (Ringly.io,2026) ⇨ providing a 90%–95% cost reduction
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	- **Time Savings Focus:** "Cutting the time small business owners spend on manual support queries by **85%**, translating to roughly 10 to 12 hours saved per week per business."

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☐ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☐ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☐ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☐ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☐ Yes ☐ No

Name	Type	License
Chatterbox Multilingual V3	Speech model and code	MIT (model card)
faster-whisper	Speech recognition code	MIT
Whisper base	Speech recognition model	MIT
PyTorch	Machine learning framework	BSD-style
FastAPI	API framework	MIT

LiveKit Agents	Voice-agent framework	Apache-2.0
Qwen/Qwen2.5-7B-Instruct	Model	Apache-2.0
Qwen/Qwen2.5-VL-7B-Instruct	Model	Apache-2.0
Faster Whisper (SYSTRAN/faster-whisper)	Code	MIT

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

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7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



October 22,2026

Architecture

